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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte(Deemed to be University)

Fourth Semester B.E. (ISE) (Credit System) Degree Examinations

Supplementary Examinations - August 2025

21IS403 – DIGITAL SYSTEM DESIGN

Duration: 3 Hours

Max. Marks:100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Unit – I		Marks	BT*	CO*	PO*
1.	a) Using K- Map find the minimized sum of product expression for the following switching function. $f(a,b,c,d)=\sum m(1,5,6,7,9,11,12,13)+dc(0,3,4)$	6	L*2	1	2
	b) With the neat circuit diagrams, prove the universality of NAND and NOR gates.	6	L1	1	1
	c) Get a minimized expression for $Y=F(A,B,C)=\sum m(1,4,6,7)$ using Quine-McClusky method and implement it using universal gates only.	8	L3	2	2
2.	a) Obtain the number $(225)_{10}$ to i) Binary ii) Octal	6	L2	1	1
	b) Find the Minimal Sum of the given Boolean Expression using K-Map: $f(a, b, c, d) = \prod M(0,1,4,5,8,9,11) + d(2,10)$	6	L2	2	2
	c) Simplify the following equations using Boolean algebra: i. $Y=A(A'+C)(A'B+C)(A'BC+C')$ ii. $Y=(A+B)(A'(B'+C'))'+A'(B+C)$	8	L2	1	1
3.	a) Given the simplified expression of a Boolean function $Y=f(A,B,C)=C'$ i. Write the truth table ii. List the min-terms iii. Obtain the simplified function using K-map method	6	L2	1	1
	b) Simplify the following Boolean equation using Boolean algebra and using only NAND gates, design the circuit for the simplified equation. $Y=(A+B)(A'C'+C)(B'+AC)'$	6	L2	1	1
	c) Simplify the Following function using Quine-McClusky method and implement it using NAND gates only. $F = \sum m(1,2,8,9,12,13,14,15)$	8	L3	2	2
Unit – II					
4.	a) Design SR flip flop with NOR latch. Explain its truth table.	6	L2	4	1
	b) Implement the function $F(A,B,C,D)=\sum m(6,7,9,10,13)+d(1,4,5,11)$ using 8:1 multiplexer (Use Entered Variable Map method)	6	L3	3	2
	c) Realize half adder and full adder with truth table, Boolean equation and circuit diagram.	8	L3	3	2
5.	a) Implement the following functions using 3:8 Decoder. $F1 = A'BC+ABC$ $F2 = \sum m(1,2,5)$ $F3 = \pi M(1,2,5,7)$	6	L3	3	2
	b) Explain with diagram how SR flip-flop is used as a switch debounce circuit.	6	L3	4	2

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- c) Implement Full Adder Carry = $\sum m(3,5,6,7)$ using i) 8:1 multiplexer ii) 4:1 Multiplexer.

8 L3 3 2

6. a) With a neat diagram illustrate the design of full adder using two half adders.
 b) With the neat logic diagram, truth table and timing diagrams, explain the working of positive edge triggered T flip-flop.
 c) Show how to convert JK flip-flop to T flip-flop with the help of state synthesis table.

6 L1 3 1

6 L2 4 1

8 L3 4 2

Unit – III

7. a) Design 3-bit Ripple down counter using T flip-flop. Also explain its truth table and timing diagram.
 b) Explain with block diagram-
 I. Serial-in Serial-out shift register
 II. Parallel-in serial-out shift register
 c) Design mod-4 synchronous counter with regular sequence (Use any flip flop).
 8. a) Design 3-bit Ripple Up counter using T flip-flop. Also explain its truth table and timing diagram.
 b) Write a note on sequence generator and sequence detector.
 c) Determine characteristic equation and excitation table of the following flip-flops
 i) SR flip-flop
 ii) JK flip-flop
 iii) D flip-flop
 iv) T flip-flop

6 L3 5 2

6 L2 5 1

8 L1 5 1

6 L1 5 1

6 L3 5 2

8 L1 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
